

Voice-Based Post-Discharge Patient Care and Monitoring System

1. Introduction

1.1 Background of the Study

Sri Lanka's healthcare system has major challenges when it comes to maintaining patient care after they leave the hospital. This is especially true in public hospitals, which frequently lack resources while serving most of the population. Once discharged, patients transition from a regulated medical setting to self-care at home. This transition is critical because difficulties, failure to adhere to medication schedules, and uncertainty about care instructions can all lead to avoidable readmissions and poor health outcomes.

Current post-discharge follow-up in Sri Lanka depends largely on nursing staff making phone calls, written discharge summaries, and scheduled clinic appointments. These methods face major challenges: staff shortages, heavy patient loads, limited phone access in rural areas, low health literacy in some groups, and wide language diversity across the island. The Ministry of Health estimates that about 15–20% of discharged patients have complications within 30 days, and many of those problems go unreported until an emergency readmission is needed.

Artificial intelligence, especially natural language processing, speech recognition, and machine learning, has shown a lot of promise in healthcare around the world. Voice-based AI can understand spoken questions, give people health information that is specific to them, pick up on worrying symptoms from how people speak, and let doctors know about possible problems. These tools could help bridge the gap between hospital discharge and community care by allowing for ongoing, scalable, and language-accessible patient monitoring.

The combination of Sri Lanka's mobile connectivity, the urgent need for better post-discharge care, and developments in AI-powered voice technologies presents an excellent opportunity to create language-inclusive, culturally appropriate solutions that can improve patient outcomes while decreasing the workload for medical professionals. To address the distinct linguistic, cultural, and infrastructural features of the local health system, this research explores how voice-based AI systems can be designed, implemented, and evaluated, especially for the Sri Lankan healthcare context.

1.2 Motivation

Many Sri Lankan patients struggle once they leave the hospital. This is particularly challenging for older patients, those living in rural regions, and those who cannot read well. Many patients become confused about their prescriptions or are unsure when their symptoms are severe enough to require medical attention. Meanwhile, medical staff are already overwhelmed and struggle to contact and check on all discharged patients.

There is limited research on AI-powered healthcare solutions specifically designed for Sri Lanka, especially when it comes to multilingual needs and regional infrastructural limitations. Discovering these issues directly demonstrated the potential benefits of voice-based AI, which would enable patients to receive assistance via straightforward phone calls regardless of their reading ability and facilitate the tracking of numerous patients simultaneously by medical professionals. The goal of this research is to design a practical tool that could enhance patient care and demonstrate how AI technology can be modified for constrained healthcare systems.

1.3 Study Objectives

Primary Objective

- Design and prototype an AI-powered voice-based system for post-discharge patient monitoring, as well as assess its feasibility and user acceptance in the healthcare environment.

Secondary Objectives

- Interviewing healthcare professionals and patients to identify significant post-discharge care requirements and obstacles.
- To create a voice interaction framework that works in Sinhala, Tamil, and English, with a focus on how people naturally talk about their health.
- To provide a functioning prototype that provides critical capabilities such as symptom checks, prescription reminders, and basic health advice via voice interactions.
- To test the prototype with a small number of discharged patients and healthcare personnel to see whether they find it useful, simple to use, and culturally suitable.
- To make recommendations on how this technology could be further enhanced and integrated into existing hospital workflows in the future.

2. Literature Review and Research Gap

2.1 Overview of Existing Studies

2.1.1 AI Conversational Agents Effectiveness Review

Milne-Ives and colleagues conducted a systematic review to evaluate the effectiveness and usability of AI conversational agents in healthcare. They explored six databases for articles published since 2008, eventually resulting in 31 studies. The methodology consisted of thorough data extraction and risk of bias evaluation using well-established methodologies. 14 chatbots (2 voice-based), 6 embodied conversational agents, and a variety of additional formats were investigated. The findings revealed generally positive evidence for effectiveness and usability, but the review also uncovered major methodological flaws. According to the quality assessment, cross-sectional studies satisfied only 50% of quality criteria on average, whereas other study designs met 42%. The evaluation concluded that, while conversational agents had potential for behavior modification, treatment assistance, and health monitoring, inadequate study design and insufficient reporting made accurately assessing their genuine utility in healthcare settings challenging. Users had conflicting qualitative judgments, particularly about trust and the lack of human empathy in AI systems [1].

2.1.2 Natural Language Processing for South Asian Languages in Healthcare

Jayawardena and his team created and tested a natural language processing system designed exclusively for Sinhala-language medical records in Sri Lanka. A collection of medical records from Sri Lankan hospitals was utilized to train and test the system's capacity to extract medical entities and interpret clinical terminology. The process included annotating medical literature, creating domain-specific word embeddings, and comparing results to gold standard annotations. The results revealed 82% accuracy in detecting medical terminology and concepts in Sinhala text. However, the study revealed substantial obstacles specific to South Asian languages, such as a scarcity of annotated medical corpora, code-switching between Sinhala and English, regional dialect variances, and a lack of standardized medical terminology in Sinhala. The system did not investigate voice-based applications because it focused only on written text processing instead of spoken language processing. The study underlined the importance of larger datasets, standardized medical terminology databases, and speech recognition capabilities for Sinhala healthcare applications, as well as the technical difficulty of creating NLP systems for low-resource languages [2].

2.1.3 Survey of Conversational Agents in Health (Montenegro et al., 2019)

Montenegro and colleagues conducted a comprehensive assessment of conversational agents in healthcare, reviewing about 4,145 papers and selecting 40 for detailed study based on PICOC (Population, Intervention, Comparison, Outcome, Context) guidelines. The study created a thorough taxonomy that classified agents based on their architecture, dialogue management methodologies, and application domains. Key findings revealed that 82% of agents used rule-based

or pattern-matching approaches rather than advanced AI, with only 15% supporting non-English languages. The review revealed text-based dialogue as the most frequent interface (used by more than 70% of agents), with multimodal interfaces (which include speech, facial recognition, and gestures) being less common. The study identified older people as a significant challenge for conversational agent creation. The study discovered considerable gaps in post-discharge monitoring applications, with most agents concentrating on appointment scheduling, general health information, or chronic illness management rather than acute postoperative care. The authors highlighted difficulties in natural language processing accuracy, connection with existing hospital systems, and obtaining suitable levels of user trust [3].

2.1.4 Acceptability of AI Chatbots in Healthcare (Nadarzynski et al., 2019)

This mixed-methods study examined patient approval of AI-led chatbot services using 29 semi-structured interviews and an online poll of 216 individuals. Thematic analysis found three major themes: knowledge of chatbots, AI hesitation, and motives for health chatbots. The quantitative results demonstrated moderate general approval (67%), although acceptance varied greatly by use case. The study discovered that acceptance was adversely correlated with weak IT skills (OR=0.32) and a dislike for communicating with computers (OR=0.77) but positively associated with perceived usefulness (OR=5.10), optimistic attitude (OR=2.71), and trustworthiness (OR=1.92). Participants were concerned about accuracy, cybersecurity, and AI's inability to display empathy. According to the findings, chatbots were more acceptable for administrative duties and obtaining anonymous health information on sensitive themes but less acceptable for serious health conditions that required human judgment and emotional support. The study underscored the importance of using user-centered techniques in intervention design to address patient concerns about privacy, accuracy, and the loss of personal connection in healthcare interactions [4].

2.1.5 Physicians' Perceptions of Healthcare Chatbots (Palanica et al., 2019)

Palanica and colleagues surveyed 100 practicing physicians across the United States to learn about their views on healthcare chatbots. The online survey asked physicians about the benefits, obstacles, and hazards of using chatbots. While physicians acknowledged administrative benefits (78% said chatbots may help with appointment scheduling, 76% with identifying health facilities, and 71% with delivering drug information), they expressed substantial concerns about clinical uses. Only 28% of physicians felt comfortable with chatbots offering diagnoses, while 64% were concerned about incorrect self-diagnosis causing patient harm. Most physicians (72%) believed chatbots could not adequately care for all patient needs and lacked the ability to display appropriate emotion. However, 76% acknowledged that chatbots could reduce their administrative workload. The study revealed a clear preference among physicians for chatbots serving as assistive tools that complement rather than replace human clinical judgment, particularly in contexts requiring complex decision-making or empathetic patient interaction [5].

2.1.6 Conversational Agents in Healthcare - Systematic Review (Laranjo et al., 2018)

Laranjo and colleagues published the first systematic review of conversational agents in healthcare that can use natural language input. Their technique consisted of searching five main databases (PubMed, Embase, CINAHL, PsycInfo, and ACM Digital Library) from inception to February 2018, retrieving 1,513 articles, and eventually incorporating 17 studies encompassing 14 different conversational agents. Conversational agents were discovered to be employed for behavior change assistance, health monitoring, and patient education. Their key findings revealed that most systems (about 80%) used limited input rather than unconstrained natural language and that the area was still in early stages, with most papers published around 2010. The analysis found significant variation in evaluation methodologies, with only one randomized controlled trial demonstrating effectiveness in decreasing depressive symptoms. The study found that while conversational bots showed potential for healthcare applications, there were significant gaps in evaluation quality, a lack of standardized testing techniques, and minimal evidence of clinical effectiveness [6].

2.1.7 Hospital Readmission Risk Prediction Models (Artetxe et al., 2018)

Artetxe and colleagues conducted a systematic assessment of predictive models for hospital readmission risk, analyzing 73 research methodologies. The review examined several machine learning and statistical approaches, including logistic regression, decision trees, neural networks, and ensemble methods. The results demonstrated that these models had modest predictive accuracy in detecting high-risk patients, ranging from 70% to 85%. Patient demographics (age, comorbidities), previous admission history, duration of stay, discharge medications, and adherence to post-discharge follow-up were all common predictors. However, the review found serious flaws in existing models. Most were created and verified using data from Western healthcare systems and may not be applicable to developing nations with diverse healthcare institutions, patient populations, and post-discharge support systems. The study found that few models included real-time post-discharge data collection or patient-reported symptoms, indicating a missed potential for early intervention. The study stressed the importance of models that can be dynamically updated with post-discharge information and tailored to various healthcare situations and resource constraints [7].

2.1.8 Patient Safety Risks with Commercial Voice Assistants (Bickmore et al., 2018)

Bickmore and colleagues examined the safety and accuracy of prominent commercial voice assistants (Siri, Alexa, and Google Assistant) in responding to health-related questions. The research technique included testing 70 typical health questions across many platforms and comparing replies to medical recommendations and expert analysis. The study discovered shockingly low accuracy rates, with voice assistants accurately answering only 55-60% of health-related questions. Unfortunately, the systems occasionally delivered possibly harmful or inaccurate responses, particularly concerning prescription combinations, emergency symptoms, and mental health emergencies. The study identified key safety vulnerabilities such as an inability to recognize medical emergencies, the supply of incorrect dose information, and the failure to promptly identify urgent symptoms. The study noted that current commercial voice assistants lack the medical

knowledge base, safety regulations, and context awareness required for accurate healthcare recommendations. The authors found that using purpose-built healthcare speech systems with sufficient clinical control and safety safeguards is preferable to relying on general-purpose consumer technologies [8].

2.2 Identification of Research Gap

Existing research on conversational AI in healthcare highlights significant gaps, particularly in language and multilingual design.

Language and Multilingual Design Gap: Most healthcare conversational agents are written in English and targeted for Western audiences [6] [3]. Only 15% of the systems surveyed support non-English languages, and none are explicitly intended for South Asian multilingual environments where patients often code-switch between Sinhala, Tamil, and English during healthcare talks. Jayawardena et al. (2020)'s Sinhala NLP research concentrated on written medical records rather than genuine spoken language, leaving voice-based healthcare applications for Sri Lankan patients untested [2]. Existing literature does not address the cultural factors of how South Asian patients express health concerns, such as communication methods, health belief systems, and familiarity with technology.

Voice-First Interface Gap: The majority of healthcare conversational agent research and development has been on text-based chat interfaces that can be accessed via smartphones [1]. However, these require data connectivity, smartphone ownership, and literacy resources that aren't always accessible in developing countries. There is a lack of research on voice-only systems that operate on entry-level mobile phones through straightforward voice calls without requiring digital literacy or internet access. According to Nadarzynski et al. (2019), this disparity is especially important for low-literate people, elderly patients, and rural populations who would most benefit from easily available voice-based healthcare support but are not eligible for text-based solutions [4].

Post-Discharge Application Gap: While there has been research on conversational agents for chronic disease management and medication reminders, their use for post-discharge patient monitoring remains insufficiently addressed [3]. Critical needs during this transition, such as wound care, detecting surgical complications, managing medications, and identifying urgent symptoms, require further study. Current readmission prediction models mainly use historical data and do not incorporate real-time symptom monitoring, missing chances for early intervention [7].

Safety and Clinical Integration Gap: Research highlights the safety and clinical integration gap in voice assistants for healthcare. Bickmore et al. (2018) found such systems provide unreliable health advice, lacking necessary safety mechanisms and clinical validation [8]. There's a lack of guidance on escalation procedures to healthcare providers and integration into hospital workflows, especially in resource-limited settings. Concerns persist about chatbot safety and the importance

of complementing clinical judgment, as noted by Palanica et al. (2019), yet frameworks for achieving this are still underdeveloped [5].

User Acceptance in Developing Countries Gap: User acceptance studies predominantly focus on developed countries [4], leaving a gap in research regarding the acceptance of AI voice systems for health monitoring in developing nations like Sri Lanka. Factors influencing adoption, such as cultural attitudes, data privacy concerns, and preferences for human interaction versus automation, require further exploration. The 67% acceptance rate identified in Western contexts may not be applicable to South Asian populations across varying demographic backgrounds.

To fill these gaps, this research will design a voice-based post-discharge tracking system for the Sri Lankan context, create a working prototype, assess whether patients and healthcare workers find it useful and acceptable, and make practical recommendations for how similar systems could be implemented in other South Asian hospitals with limited resources.

3. Research Problem and Questions

3.1 Main Research Problem

How can a voice-based AI system be built and prototyped to efficiently monitor post-discharge patients in Sri Lankan hospitals while remaining accessible, culturally suitable, and feasible with current resource constraints?

3.2 Specific Research Questions

1. What specific issues or sub-problems will the study address?
 - Language barriers prevent effective post-discharge communication (Sinhala, Tamil, English)
 - Low health literacy makes it harder for patients to understand discharge instructions.
 - Limited access to smartphones and internet connectivity in rural areas.
 - The shortage of healthcare workers requires manual follow-up calls. (time-consuming, unsustainable)
 - Delayed diagnosis of post-discharge concerns, resulting in avoidable readmissions
 - Lack of culturally relevant health technology intended for Sri Lankan people.
 - Insufficient monitoring of medication adherence and wound care following hospital discharge.
2. How will your approach or system solve these problems?
 - Voice-based interface accessible via typical phone calls, requiring no smartphone, applications, or internet connection.
 - Multilingual support enables patients to naturally speak in their choice language (Sinhala, Tamil, or English).
 - Simple conversational design, with clear inquiries. Suitable for users with different literacy levels.
 - Automated daily or planned check-ins reduce workload for medical staff.
 - Rule-based symptom detection system identifies warning indicators and red flags early.
 - Alert and notification system to quickly inform healthcare providers if patients report concerning symptoms.
 - A structured approach for assessing drug compliance, wound care, pain levels, and common consequences.
 - Clear health advice and reassurance for typical post-discharge concerns
3. Which techniques, datasets, or models will you use?
 - Speech recognition APIs and services for Sinhala, Tamil, and English languages (e.g., Google Speech-to-Text, Azure Speech Services)
 - Natural language processing approaches to comprehend patient comments, extract keywords, and identify intent

- A rule-based dialogue management system for systematic health assessments and conversation flow
- Symptom classification algorithms use preset medical criteria to identify high-risk conditions that require prompt intervention.
- Voice application development platforms such as Twilio for voice calls and Google Dialogflow for conversation design
- user-centered design process, including semi-structured interviews and focus groups with discharged patients and healthcare personnel.
- Iterative design-test-refine cycles are used to create prototypes depending on user feedback.
- Interviews were conducted to gather qualitative input on user experiences, trust levels, and acceptance criteria.
- Post-discharge patient data and medical practices from Sri Lankan state hospitals (with proper ethical approval and informed consent)
- Post-prototype evaluation surveys measure acceptance rates, ease of use, trust levels, and perceived usefulness.
- Frameworks for comparing prototype performance to existing manual follow-up techniques

4. Research Strategy and Methodology

4.1 Research Strategy

This research applies to a Design Science Research (DSR) approach, which is best suited for creating and assessing modern technology solutions that address real-world issues.

Justification for Design Science Research:

1. Artifact Creation Focus: Rather than just studying current systems, DSR is specifically made for research that creates new IT artifacts. DSR offers the proper methodological framework for the methodical creation, testing, and assessment of this technical solution because the aim of this project is to design and construct a functional voice-based post-discharge tracking system prototype.

2. Problem-Solving Orientation: DSR places a significant value on using creative solutions to address real-world issues. The practical issue of insufficient post-discharge monitoring in Sri Lankan hospitals is directly addressed by this research. Instead of staying simply theoretical, the DSR approach guarantees that the research creates practical results that may be applied in real healthcare settings.

3. Iterative Development and Refinement: DSR encourages iterative cycles of design, evaluation, and improvement based on user feedback, which is essential to creating a user-centered voice system. This method involves evaluating patients and healthcare personnel, making improvements depending on their feedback, and going through numerous iterations to successfully meet genuine requirements in local settings.

4. Rigorous Evaluation Framework: Rigorous evaluation in Design Science Research (DSR) requires a systematic assessment of artifacts based on criteria such as usability, acceptability, feasibility, and effectiveness, which is critical for determining the prototype's functionality for Sri Lankan users and viability in resource-constrained hospital settings.

5. Context-Specific and Practical Relevance: Design Science Research (DSR) addresses the need to adjust technical solutions to unique situations. This research focuses on the distinct language, cultural, and infrastructural aspects of Sri Lankan healthcare, ensuring that the generated prototype and recommendations are relevant and applicable to similar resource-constrained South Asian situations rather than presenting generic solutions.

4.2 Research Methodology Framework

This research follows Peffers' Design Science Research Methodology (DSRM), which outlines a systematic six-stage framework for conducting design science research.

1. Problem Identification

This stage includes determining and specifying the exact issue that the research will address. Activities involve conducting an extensive literature review to identify existing gaps in post-discharge monitoring systems, assessing problems in Sri Lankan healthcare, and establishing the need for accessible, multilingual voice-based patient tracking. This stage emphasizes the need and relevance of creating a voice-based system that is specifically designed for resource-constrained contexts.

2. Define Objectives

This stage defines specific objectives for the voice-based system, such as the creation of a multilingual interface, automatic post-discharge check-ins, symptom detection, alert mechanisms for healthcare providers, and accessibility for low-literacy users. These objectives are driven by insights acquired from user interviews and surveys, ensuring that they satisfy the actual needs of patients and healthcare staff while also providing evaluation targets for the outcome.

3. Design and Development

This stage includes creating the voice-based post-discharge tracking system prototype. Key activities include designing the system architecture and conversation flow, selecting appropriate technologies (speech recognition APIs, NLP tools, and voice platforms), developing multilingual capabilities for Sinhala, Tamil, and English, creating dialogue scripts and symptom assessment protocols, implementing rule-based algorithms for identifying concerning symptoms, developing an alert system for healthcare providers, and integrating safety mechanisms. The development is guided by user-centered design principles, with iterative refinement based on feedback from small-scale testing with the intended users.

4. Demonstration

This stage involves showing how the prototype works and proving that it can effectively solve the defined problem. The device is installed in a monitored environment, and discharged patients utilize it for post-discharge monitoring for a set length of time (often 7-14 days). Healthcare personnel use the alert system and examine patient responses. Real-world events are replicated, such as drug requests, illness reporting, and emergency circumstances. Documentation is generated to demonstrate how the system functions across multiple use cases, languages, and patient types. This stage demonstrates the technical viability and practical functionality of the voice-based system prior to a thorough review.

5. Evaluation

This step thoroughly examines the prototype against the objectives set in step 2. Multiple evaluation methods are used, including usability testing with standardized instruments such as the System Usability Scale (SUS), post-use surveys measuring user satisfaction and acceptance, interviews with patients and healthcare workers to collect qualitative feedback, analysis of system performance metrics (call completion rates, symptom detection accuracy, and response times), and assessment of technical challenges encountered during demonstration. The review examines whether the system is useable, acceptable, accurate, feasible to implement, and culturally suitable. The findings reveal strengths, flaws, and places for improvement. If substantial flaws are discovered, the procedure may return to Stage 3 for refining.

6. Communication

The final stage entails disseminating research findings to appropriate audiences. Activities include documenting the problem, the designed solution, the development process, and the evaluation results in the final research report, preparing academic publications and conference presentations, creating practical implementation guidelines and making recommendations for future research and development. The communication guarantees that the knowledge obtained by this study helps both academic understanding and practical improvements in post-discharge care in resource-constrained settings.

4.3 Data Collection Methods

This research will use both primary and secondary data collection methods.

Primary Data Collection

Interviews: Semi-structured interviews will be conducted with discharged patients as well as healthcare staff. Interviews will explore existing post-discharge difficulties, communication preferences, technological comfort levels, and system requirements.

Surveys: Multiple surveys will be administered, including a baseline survey for discharged patients assessing current experiences, technology access, and language preferences; a healthcare worker survey examining workload and attitudes towards technology; and post-demonstration usability surveys using the System Usability Scale (SUS) to measure ease of use, satisfaction, and acceptance among prototype users.

Observation: Healthcare staff conducting routine follow-up calls and patients interacting with the prototype system will be observed closely. During system contact, field notes will be used to record discussion patterns, obstacles faced, technical concerns, and user behavior.

Voice Recordings: With participant approval, sample interactions between patients and the system will be recorded and transcribed

Secondary Data Collection.

Medical Guidelines: Clinical practice guidelines and medical literature from the Sri Lankan Ministry of Health, WHO standards, and medical textbooks will be used to inform symptom assessment techniques and safety measures.

Hospital Records: Aggregated, anonymized data from partner institutions, such as readmission rates, common complications, and patient demographics, will give context and benchmark measures. Access will necessitate explicit agreements and ethical approval.

Existing research: Published datasets and benchmarks for voice-based healthcare systems will help define design decisions and performance expectations.

4.4 Data Preprocessing / Preparation

Detail the preprocessing or data cleaning steps (e.g., normalization, augmentation, feature extraction).

Qualitative Data Preparation (Interviews, Observations)

Audio recordings will be transcribed verbatim in their native language (Sinhala, Tamil, or English), then translated to English for analysis, while the original versions will be kept for cultural context.

Transcripts will be examined for accuracy, all personally identifiable information (names, addresses, and phone numbers) will be removed, and irrelevant content, such as long pauses, will be removed.

Key themes such as user demands, technical barriers, language preferences, trust problems, and desired features will be identified through a systematic coding of processed material. These themes will be used to drive system design decisions.

Survey Data Preparation

Survey replies will be reviewed for completeness and consistency, and any incomplete or invalid responses will be removed, as will duplicate entries.

Text entries will be standardized for consistency, open-ended responses will be classified thematically, and rating scales will be normalized as needed for statistical analysis.

To evaluate, composite scores such as System Usability Scale (SUS) totals will be derived from individual item replies.

Voice Recording Preparation

Recorded interactions will be processed to remove background noise, adjust volume levels, and improve overall audio quality to facilitate analysis.

recordings will be transcribed into text, with annotations indicating the language used (Sinhala, Tamil, English, or mixed), clarifying vocabulary, and flagging areas where the system struggled to understand users.

Patients' common health-related words, local medical terms, and colloquial expressions will be recognized and gathered into a reference vocabulary list to help improve system conversation design.

Final Data Preparation

Data from many sources (interviews, surveys, and observations) will be merged to provide comprehensive insights and validate findings across several datasets.

All preprocessed data will be finalized to guarantee that no information that might be used to identify participants remains.

Clean data will be kept in secure, well-organized databases, with comprehensive documentation of the pretreatment processes used, ready for analysis and system improvement.

4.5 Model Development / Implementation

System Architecture

The system will have a modular architecture with four major components: the voice interface layer, which handles phone calls and speech processing; the dialogue management layer, which controls conversation flow; the symptom assessment layer, which evaluates patient responses, and the alert notification layer, which alerts healthcare providers to concerning cases. These components will work together to enable seamless, automated patient monitoring.

➤ The Voice Interface Component

Speech Recognition: The system will incorporate speech-to-text APIs such as Google Cloud Speech-to-Text and Azure Speech Services, which support Sinhala, Tamil, and English. These APIs will transform patient voice input to text for processing.

Text-to-Speech: Text-to-speech engines translate system responses into natural-sounding voice output in the patient's preferred language. The system will employ appropriate speech parameters (gender, speed, and tone) for healthcare communication.

Telephony Integration: Voice call capability will be implemented utilizing platforms such as Twilio or similar services that allow automated phone calls to patients' mobile or landline numbers without requiring internet connection on the patient's end.

➤ Dialogue Management System

Conversation Flow Design: A rule-based dialogue management system will be created to direct talks along structured assessment paths. The flow will comprise greeting and language selection, medication compliance checks, wound care and pain assessment, symptom inquiry with yes/no and open-ended questions, health advice provision, and call completion with next check-in scheduling.

Context Management: Throughout each call, the system will preserve conversation context by remembering prior responses within the same conversation, noting which questions have been asked, and tailoring follow-up questions to patient responses.

Multilingual Support: Different dialogue scripts will be prepared for Sinhala, Tamil, and English, with culturally appropriate greetings, basic, clear health questions appropriate for varied literacy levels, and natural language based on how Sri Lankans generally address health concerns.

➤ Natural Language Processing Component

Intent Recognition: Simple NLP algorithms can determine if a patient is reporting a symptom, asking a question, confirming drug adherence, or expressing confusion.

Keyword Extraction: The system will extract pertinent medical keywords from patient responses, including symptom terms (pain, fever, bleeding, swelling), severity indications (severe, mild, worsening), and time-related information (since yesterday, three days).

Sentiment Analysis: Basic sentiment detection will detect emotional signs in patient replies such as discomfort, confusion, or worry, which may indicate the need for human intervention even if symptoms appear moderate.

➤ Symptom Assessment and Risk Stratification Algorithm

Rule-Based Symptom Evaluation: A decision-tree algorithm compares reported symptoms to preset clinical criteria. Rules will be developed in collaboration with healthcare professionals and based on medical guidelines. Examples include a high fever ($>38.5^{\circ}\text{C}$), extensive bleeding or wound discharge, intense pain that cannot be eased by medication, and symptoms of infection (redness, warmth, swelling).

Risk Scoring System: Each reported symptom will be awarded a risk score based on severity; the combination of numerous symptoms increases the overall risk score, and patients who exceed predetermined risk thresholds will get healthcare provider warnings.

Escalation Logic: The algorithm will select suitable responses: low-risk instances will receive automated health advice and reassurance, medium-risk cases will create alerts for nurse review within 24 hours, and high-risk cases will generate immediate alerts to on-call medical staff.

➤ Alert and Notification System

Healthcare Provider Dashboard: A simple web-based interface that displays patient alerts organized by priority level, shows patient call history and reported symptoms, allows physicians to mark alerts as reviewed or resolved, and allows them to contact patients directly.

Notification Mechanisms: Alerts will be given via a variety of channels, including SMS messages to designated healthcare staff, email notifications with patient summaries, and in-app notifications for healthcare workers who utilize a mobile application.

Alert Content: Notifications will include the patient identification (anonymized code), risk level and priority, important symptoms reported, time of call, and recommended action depending on symptom severity.

➤ Data Storage and Security

Database Design: A secure database will contain patient profiles with anonymous IDs, call history and transcripts, reported symptoms and answers, alert records, and system performance metrics.

Security measures: Security measures will include data encryption and transfer, access restrictions that limit who can view patient information, regular backups to prevent data loss, and compliance with healthcare data privacy legislation.

➤ Technology Stack

The technology stack includes Python for backend development and NLP and JavaScript for web-based dashboards.

Frameworks and libraries include natural language processing libraries for text analysis, dialogue management frameworks for conversation control, and web frameworks (such as Flask and Django) for dashboard creation.

Cloud services include cloud-based speech recognition APIs, cloud hosting for system setup and scalability, and safe data storage via databases.

Implementation Approach

Iterative Development: The system will be built in stages, beginning with a minimal viable prototype with basic conversation flow in one language, progressing to multilingual support and refined dialogue, implementing symptom assessment and alert features, and finally integrating all components for comprehensive testing.

User-Centered Refinement: Following each development iteration, feedback from small-scale testing with patients and healthcare staff will be used to fine-tune dialog scripts, increase voice recognition accuracy, and change risk assessment thresholds.

This modular development strategy allows each component to be tested, developed, and improved individually prior to complete system integration, resulting in a reliable and user-friendly voice-based post-discharge tracking system.

4.6 Evaluation

The voice-based post-discharge tracking system will be evaluated using several metrics and testing methodologies to measure its technical performance, usability, acceptance, and feasibility of implementation.

Technical Performance Evaluation

Speech Recognition Accuracy: Determines how well the system understands what patients say in Sinhala, Tamil, and English. Percentage of words properly identified. Target: 85% or higher.

Call Completion Rate: The percentage of calls that were completed successfully, with no technical issues or patients hanging up. The formula is $(\text{completed calls} / \text{total attempted calls}) \times 100$. Target: 80% or higher.

Response Time: Response time refers to how quickly the system replies after patients talk. It should take no more than 3-5 seconds to feel natural.

Symptom Detection Accuracy: How well the system recognizes and categorizes reported symptoms. Precision (correct alerts / all alarms given) and recall (correct alerts / all actual problems) are used to get this value. This demonstrates whether the system detects serious problems without generating too many false alerts.

Usability Evaluation

System Usability Scale (SUS): A standard 10-question survey administered to patients after they have used the system. Scores range from 0 to 100, with scores above 68 considered good and scores above 80 considered exceptional.

Task Success Rate: The percentage of users who can successfully complete a whole health check-in call. Target: 90% or greater.

Error Rate: The number of times a user becomes confused, needs to repeat answers, or has difficulty understanding questions. Lower is preferable.

Call Duration: Call duration refers to how long each check-in call takes. Should be efficient (5-10 minutes) without being rushed.

User Acceptance Evaluation

Acceptance Rate: The percentage of patients who accept to use the system when it is presented.

Continued Use Rate: The proportion of patients who continue to use the system during their monitoring period against those who cease.

Trust and contentment Surveys: Questionnaires that ask patients about their trust in the system, level of comfort, overall satisfaction, and preference for it over typical phone conversations.

Clinical Usefulness Evaluation

Early Detection: Monitor how many issues the system detects earlier than existing procedures.

Provider Response: Measure how quickly healthcare staff review and respond to system notifications.

Testing Approach

Pilot Testing Phase: The evaluation will be carried out as a pilot study with released patients over a 7–14-day period following discharge. Patients will engage with the system on a daily or biweekly basis, and outcome data will be collected continuously.

A/B Testing: If resources are available, different conversation design techniques or language variations will be tried with different user groups to determine the most effective design options.

Iterative Evaluation: The findings of the initial testing will be used to develop the system, which will then be tested again to verify the improvements. This loop repeats until the performance meets the stated thresholds.

Successful Criteria

The system will be considered successful if it achieves speech recognition accuracy greater than 85% across all languages, a SUS score greater than 68 (preferably greater than 80), a call

completion rate greater than 80%, a user satisfaction rating greater than 3.5 out of 5, symptom detection accuracy (F1-score) greater than 75%, and positive qualitative feedback demonstrating cultural appropriateness and trust.

4.7 Ethical considerations

Detail the ethical requirements of the research

This study involves human participants and sensitive health information, which requires careful adherence to legal requirements to protect participant rights, privacy, and well-being.

➤ Informed consent

Voluntary Participation: All participants (patients, caregivers, and healthcare staff) will be notified that their participation is entirely voluntary, and they may withdraw at any moment without jeopardizing their treatment or employment.

Comprehensive Information: Participants will be given clear explanations in their preferred language (Sinhala, Tamil, or English) about the research purpose, what their participation entails, how long it will take, what data will be collected, how it will be used and stored, potential risks and benefits, and their right to withdraw.

Written Consent: Participants will sign consent papers after asking questions and receiving appropriate responses. Consent for patients with limited literacy will be sought verbally in the presence of a witness and audio-recorded with authorization.

Separate Consents: Different consent forms will be used for interviews, surveys, voice recording, and system usage, allowing participants to pick which activities they agree to.

➤ Privacy and Confidentiality

Data Anonymization: All personally identifiable information (names, addresses, phone numbers, and national identification numbers) will be deleted from data and replaced with coded IDs. Solely the researcher will keep a secure master list of codes and identities, which will be available solely for follow-up purposes.

Secure Data Storage: All acquired data will be kept secure on password-protected devices, encrypted cloud storage with limited access, and locked physical storage for paper documents. Access will be restricted to the researcher and supervisor only.

Confidential Reporting: Research findings will be reported aggregately, without identifying individual identities. Quotes from interviewees will be anonymized, with any potentially identifying information deleted.

Voice Recording Protection: Due to their sensitive nature, audio recordings will be held separately from identifying information, erased after transcription unless participants consent to longer retention, and never shared outside of research supervision.

➤ **Data Handling and Retention**

Limited Retention: Data will be kept only as long as necessary for research purposes. A defined data retention timeline will specify when various categories of data will be securely deleted

Transparent Data Use: Participants will be notified that anonymized data may be used for academic papers, presentations, or shared with other researchers, but will never contain personally identifiable information.

Right to Access: Participants have the right to seek access to their own data, revisions if the data is incorrect, and deletion of their data before analysis is completed.

➤ **Institutional Approvals**

Hospital Approval: Formal approval will be obtained from the administration of participating hospitals before recruiting patients or healthcare staff.

4.8 Limitations of the study

There are various limitations to this research that may affect the scope and generalizability of its findings.

➤ **Sample and Scope Limitations**

Due to limited resources and time, the study will only be for limited patients for prototype testing and survey responders. This small sample size may not accurately reflect Sri Lanka's diverse patient population, which includes people of all ages, regions, and backgrounds.

The study will take place in only a few hospitals, which may not reflect all sorts of healthcare institutions in Sri Lanka. The findings may not apply to all rural hospitals, teaching hospitals, or private facilities.

➤ **Technical Limitations**

The current technology for Sinhala and Tamil speech recognition is less advanced than English. Regional accents and dialects may influence how well the system recognizes various users.

Due to time constraints, the system will employ basic algorithmic approaches rather than advanced machine learning. This inhibits the system's capacity to handle more sophisticated, nuanced talks than established patterns.

➤ **Research Design Limitations**

Much of the evaluation is based on surveys and interviews, which may be biased because participants provide socially desired answers rather than entirely honest input.

Due to the short trial period and small sample size, actual clinical outcomes such as reduced readmissions or improved health cannot be measured. The study can only measure proxies such as user satisfaction and early symptom detection.

➤ **Resource and Time Constraints**

The scope of data collecting and testing may be limited by gaining access to hospitals and collaborating with busy medical staff.

Despite these constraints, this research will give valuable insights about the feasibility, acceptability, and possible usefulness of voice-based post-discharge tracking systems in Sri Lanka. The findings will provide the framework for future study and development in this field, and the limits identified will inform recommendations for larger-scale studies and implementations.

4. Timeline

Phase	Timeline	Key Activities	Milestones
Problem Identification & Literature Review	(2 months)	Complete and comprehensive literature review. Identify research gaps and provide a theoretical framework. Obtain ethical approval. Ensure hospital approval and access permissions.	Ethics approval obtained literature review completed
Requirements Gathering & Objectives Definition	(1 month)	Use data collection techniques. Examine the current follow-up procedures after discharge. Specify the system's needs and success standards.	System requirements document Defined objectives and evaluation criteria
System Design & Prototype Development	(2 month)	Design the system architecture and communication flows. Choose Speech Recognition APIs and Voice Platforms Create multilingual dialogue scripts (Sinhala, Tamil, and English).	System architecture document Conversation flow diagrams Working prototype of voice-based system Technical documentation Internal testing results

		Develop symptom assessment protocols. Implement rule-based algorithms. Create an alert and notification system. Conduct internal tests and refinements.	
Demonstration & Pilot Testing	(1 month)	Monitor technical performance and user interactions Document system performance	Technical performance report
Evaluation & Feedback Collection	(1 month)	post-use surveys (SUS and acceptance questionnaires) Collect qualitative feedback Analyze system metrics and technical data	User feedback summaries
Data Analysis & System Refinement	(2 month)	Transcribe and analyze data Perform statistical analysis of survey data Evaluate systems against defined metrics Conduct thematic analysis Identify key findings	Data analysis report Statistical analysis results Evaluation findings Identified themes and patterns Refined prototype (if applicable)

		and improvement areas Make final system refinements if needed	
Report Writing & Documentation	(1 month)	Write comprehensive research report Develop implementation guidelines Create technical documentation Prepare academic publications	Complete research report Implementation guidelines Technical documentation Draft publications
Finalization	(1 month)	Finalize and submit research report Prepare presentation for research defense Complete thesis submission	Final thesis submitted and research defense completed

References

- [1] C. d. C. E. L. M. H. S. N. d. P. G. M. E. N. a. E. M. M. Milne-Ives, "The effectiveness of artificial intelligence conversational agents in health care: Systematic review," *J. Med. Internet Res.*, vol. 22, October 2020.
- [2] N. S. a. S. R. N. Jayawardena, "Natural language processing for Sinhala: Applications and challenges," in *Proc. Int. Conf. Adv. ICT Emerg. Reg. (ICter)*, pp. 177-182, 2020.
- [3] C. A. d. C. a. R. d. R. R. . L. Z. Montenegro, "Survey of conversational agents in health," *Expert Syst. Appl.*, vol. 129, pp. 56-67, September 2019.
- [4] O. M. A. C. a. D. R. T. Nadarzynski, "Acceptability of artificial intelligence (AI)-led chatbot services in healthcare: A mixed-methods study," *Digit. Health*, vol. 5, 2019.
- [5] P. F. A. T. M. L. a. Y. F. A. Palanica, "Physicians' perceptions of chatbots in health care: Cross-sectional web-based survey," *J. Med. Internet Res.*, vol. 21, April 2019.
- [6] L. L. e. al., "Conversational agents in healthcare: A systematic review," *J. Am. Med. Inform. Assoc.*, vol. 25, pp. 1248-1258, September 2018.
- [7] A. B. a. M. G. A. Artetxe, "Predictive models for hospital readmission risk: A systematic review of methods," *Comput. Methods Programs Biomed.*, vol. 164, pp. 49-64, September 2018.
- [8] H. T. S. O. T. K. O. R. A. N. M. R. a. R. C. T. W. Bickmore, "Patient and consumer safety risks when using conversational assistants for medical information: An observational study of Siri, Alexa, and Google Assistant," *J. Med. Internet Res.*, vol. 20, September 2018.